4.0A STANDARD RECOVERY BRIDGE RECTIFIER

Product Summary (@T_A = +25°C)

V_{RRM} (V)	I_O MAX (A)	V_F MAX (V)	I_R MAX (µA)
400, 600, 800, 1000	4	1.1	5

Description and Applications

Suitable for AC to DC bridge full wave rectification for AC/DC power supplies, LED lighting, home appliances, office equipment, and telecommunication applications.

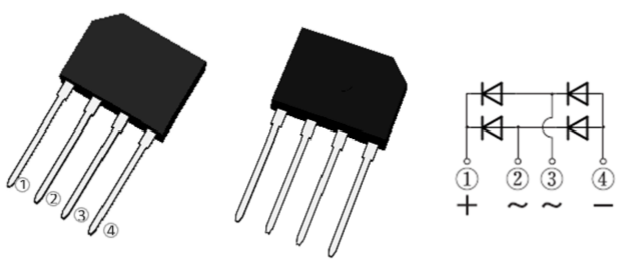
Features and Benefits

- Glass Passivated Die Construction
- Rating to 1000V PRV
- Low Reverse Leakage Current
- Surge Overload Rating to 130A Peak
- Ideal for Printed Circuit Board Applications
- UL Recognized File # E95060
- **Lead-Free Finish; RoHS Compliant (Notes 1 & 2)**
- **For automotive applications requiring specific change control (i.e. parts qualified to AEC-Q100/101/104/200, PPAP capable, and manufactured in IATF 16949 certified facilities), please contact us or your local Diodes representative.** **https://www.diodes.com/quality/product-definitions/**

Mechanical Data

- Package: KBP
- Package Material: Molded Plastic. UL Flammability Classification Rating 94V-0
- Terminals: Finish – Bright Tin, Plated Leads. Solderable per MIL-STD-202, Method 208 ⓔ3
- Polarity: Marked on Body
- Weight: 1.52 grams (Approximate)

KBP

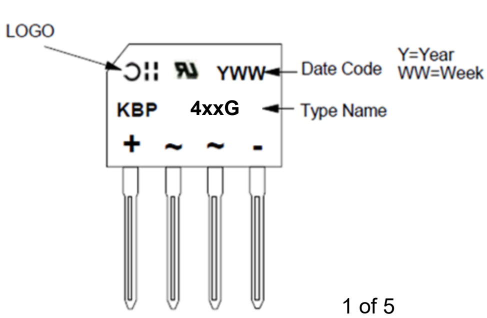


Ordering Information (Note 3)

Orderable Part Number	Package	Packing	
		Qty.	Carrier
KBP404G	KBP	35 Pieces	Tube
KBP406G	KBP	35 Pieces	Tube
KBP408G	KBP	35 Pieces	Tube
KBP410G	KBP	35 Pieces	Tube

Notes: 1. EU Directive 2002/95/EC (RoHS), 2011/65/EU (RoHS 2) & 2015/863/EU (RoHS 3) compliant. All applicable RoHS exemptions applied.
2. See https://www.diodes.com/quality/lead-free/ for more information about Diodes Incorporated's definitions of Halogen- and Antimony-free, "Green" and Lead-free.
3. For packaging details, go to our website at https://www.diodes.com/design/support/packaging/diodes-packaging/.

Marking Information



Maximum Ratings (@T_A = +25°C, unless otherwise specified.)

Single phase, half wave, 60Hz, resistive or inductive load.
For capacitive load, derate current by 20%.

Characteristic	Symbol	KBP404G	KBP406G	KBP408G	KBP410G	Unit
Peak Repetitive Reverse Voltage Working Peak Reverse Voltage DC Blocking Voltage	V_{RRM} V_{RWM} V_{RM}	400	600	800	1000	V
RMS Reverse Voltage	$V_{R(RMS)}$	280	420	560	700	V
Average Rectified Output Current @T_C = +105°C (With Heatsink) (Without Heatsink)	I_O	4.0 2.0				A
Non-Repetitive Peak Forward Surge Current 8.3ms Single Half Sine Wave Superimposed on Rated Load	I_{FSM}	130				A
Non-Repetitive Peak Forward Surge Current 1.0ms Single Half Sine Wave Superimposed on Rated Load	I_{FSM}	260				A
I^2t Rating for Fusing (3ms ≦ t ≦ 8.3ms)	I^2t	50				A^2s

Thermal Characteristics

Characteristic	Symbol	Value	Unit
Typical Thermal Resistance, Junction to Case (Note 4)	$R_{\theta JC}$	6	°C/W
Typical Thermal Resistance, Junction to Lead (Note 4)	$R_{\theta JL}$	8	°C/W
Typical Thermal Resistance, Junction to Ambient (Note 4)	$R_{\theta JA}$	15	°C/W
Typical Thermal Resistance, Junction to Case (Note 5)	$R_{\theta JC}$	14	°C/W
Typical Thermal Resistance, Junction to Lead (Note 5)	$R_{\theta JL}$	20	°C/W
Typical Thermal Resistance, Junction to Ambient (Note 5)	$R_{\theta JA}$	40	°C/W
Operating and Storage Temperature Range	T_J, T_{STG}	-55 to +150	°C

Electrical Characteristics (@T_A = +25°C, unless otherwise specified.)

Characteristic	Symbol	Min		Typ	Max	Unit	Test Condition
Reverse Breakdown Voltage (Note 6)	$V_{(BR)R}$	1,000	KBP410G	—	—	V	I_R = 5µA
		800	KBP408G				
		600	KBP406G				
		400	KBP404G				
Forward Voltage Drop per Element	V_F	—		0.94	1.1	V	I_F = 4.0A, T_J = +25°C
Leakage Current (Note 6)	I_R	—		— —	5 500	µA	V_R = 1000V, T_J = +25°C V_R = 1000V, T_J = +125°C
Total Capacitance per Element	C_T	—		40	—	pF	V_R = 4.0V_{DC}, f = 1MHz

Notes:
4. Thermal resistance per element. Device mounted on 75mm x 75mm x 1.6mm Cu plate heatsink.
5. Thermal resistance per element without heatsink.
6. Short duration pulse test used to minimize self-heating effect.

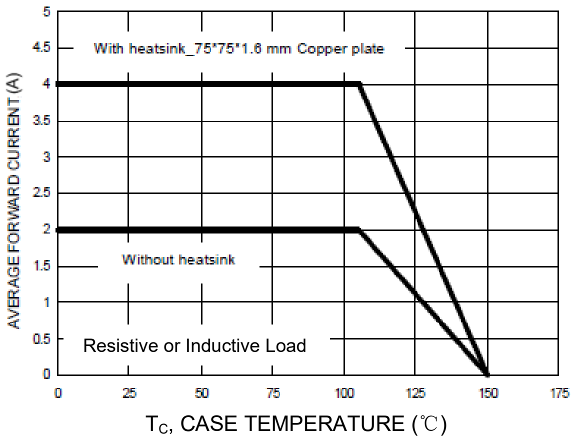


Figure 1. Forward Current Derating Curve

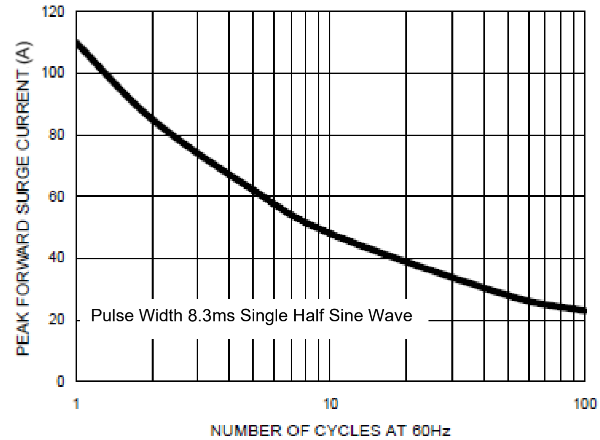


Figure 2. Maximum Non-Repetitive Surge Current

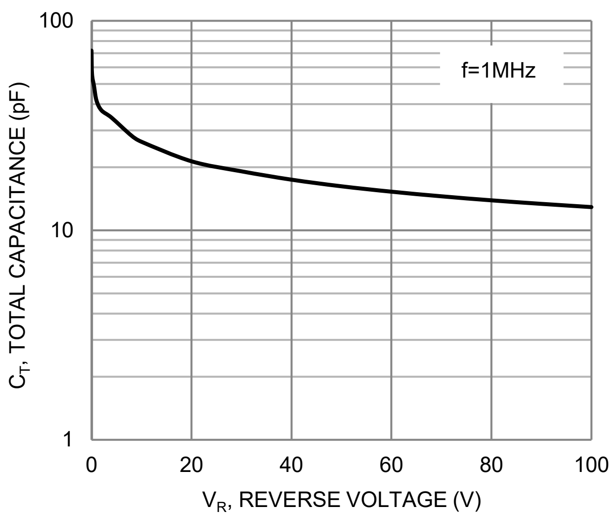


Figure 3. Typical Total Capacitance (Per Element)

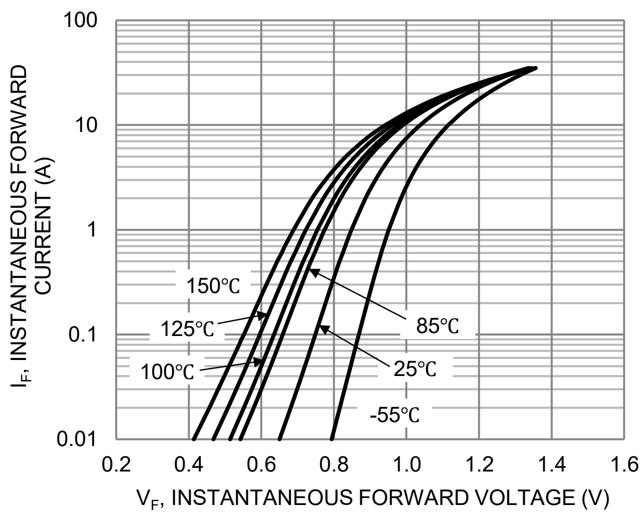


Figure 4. Typical Forward Characteristics

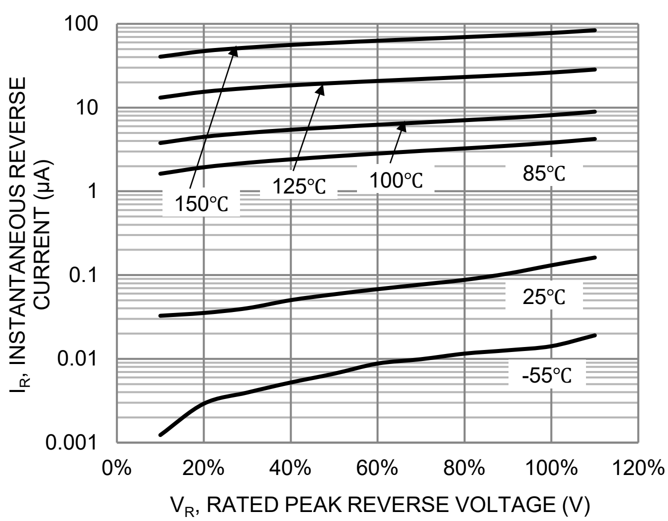


Figure 5. Typical Reverse Characteristics

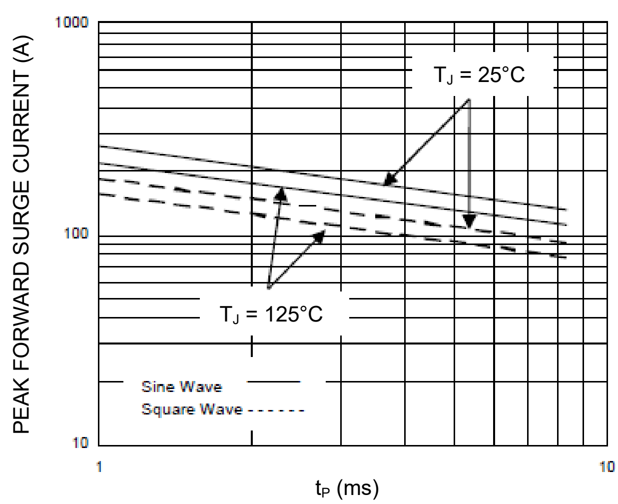
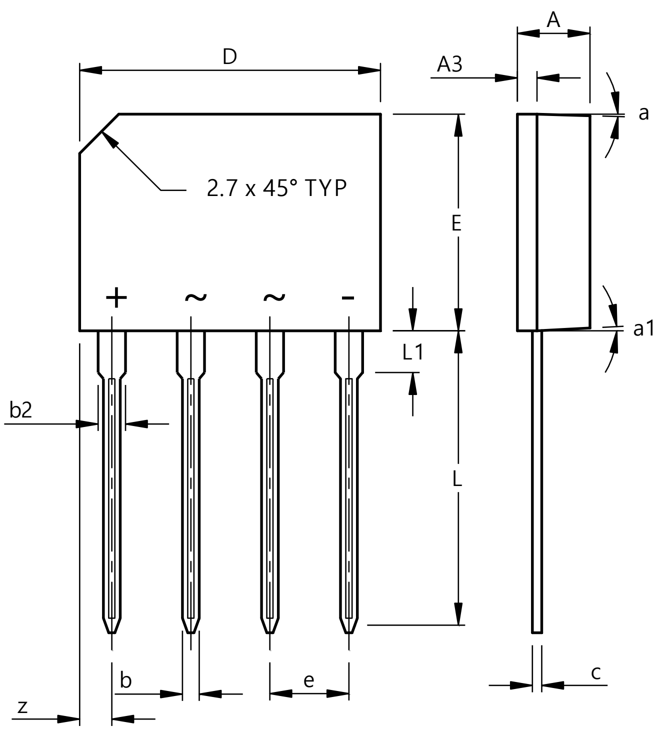


Figure 6. Non-Repetitive Surge Current

Package Outline Dimensions

Please see http://www.diodes.com/package-outlines.html for the latest version.

KBP



KBP			
Dim	**Min**	**Max**	**Typ**
A	3.35	3.65	-
A3	0.80	1.10	-
b	0.76	0.86	-
b2	1.22	1.42	-
c	0.35	0.55	-
D	14.25	14.75	-
E	10.20	10.60	-
e	3.56	4.06	-
L	14.25	14.73	-
L1	1.80	2.20	-
z	1.40	1.70	-
a	-	-	3°
a1	-	-	2°
All Dimensions in mm			

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